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Technical paper

A Multi-Agent and synergistic Knowledge Graph retrieval-augmented generation framework for intelligent maintenance☆,☆☆

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ABSTRACT

As industrial equipment is increasingly applied in manufacturing, it is necessary to adopt advanced maintenance techniques to ensure stable equipment operation. Maintenance decision support is essential for industrial equipment, but information fragmentation and complex reasoning hinder its progress. To address these challenges, we propose MAKG (Multi-Agent and synergistic Knowledge Graph), a framework that synergizes Retrieval-Augmented Generation (RAG) with a multi-agent system for intelligent industrial equipment maintenance. The MAKG framework uses these Large Language Models (LLM) agents to work together, drawing information from a diverse knowledge base to produce coordinated responses to equipment faults. Our method focuses on fine-tuning the embedding models with domain-specific data. This approach avoids the high computational cost of re-training the entire LLM and improves performance in the specific industrial context. In the experimental study, the MAKG framework was validated using real-world industrial operation and maintenance data, achieving a reasoning accuracy of 90.1% and demonstrating its potential for intelligent maintenance of industrial equipment. Code is available at https://github.com/LAM2360/MAKG_demo/tree/master.

1. Introduction

The rapid development of industry highlights the critical role industrial equipment plays in modern manufacturing, which greatly enhances production efficiency and flexibility. However, this progress faces increasing stringent demands on the reliability and safety of industrial machinery [1]. Intelligent maintenance support are often complex and long-lasting, resulting in considerable downtime and maintenance costs for the manufacturer. As a result, efficient and accurate maintenance knowledge reasoning are a key challenge for modern industry with important practical implications to increase productivity and reduce operational costs [2,3].

With the evolution of technology, intelligent maintenance methodologies have turned the focus to model based [4–6], data-driven [7–9], knowledge-based [10,11] approaches. Research on the creation of specialized domain knowledge reasoning tools has been prompted by the rapid progress in large language model (LLM) technology [11,12]. Existing studies predominantly employ LLM fine-tuning techniques [13–15] to incorporate fault record information into the model, thereby enhancing inference capabilities. However, LLMs show strong performance in general tasks, but their effectiveness in intelligent maintenance is restricted by significant limitations. The main issue is that LLMs are highly data- and computation-intensive. However, the study of LLM in intelligent maintenance is still in its infancy, where data is scarce and fine-tuning is hard to achieve satisfactory performance [16]. Furthermore, intelligent maintenance requires extremely high reasoning accuracy. In contrast, LLMs are prone to hallucinations, lack sufficient specialization, and have unknown knowledge sources [17]. As a result, although LLMs can be applied in many areas, they are still unable to perform intelligent maintenance due to these significant limitations.

Recently, retrieval-augmented generation (RAG) methodology has emerged as a knowledge-based approach to intelligent maintenance [18,19]. It automatically retrieves relevant corpora, including maintenance history and expert experience, based on fault-related queries, analyzes the issues, and generates solutions. It has shown great promise compared to LLM fine tuning techniques. In efficient diagnosis and

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rapid response, RAG retrieves reliable fault information from domain knowledge bases and provides targeted responses based on the problem context, which provides unique benefits [20]. However, current RAG systems face several critical limitations that hinder their development. Firstly, many methods rely on simplistic data representations, which restrict the ability to understand and retrieve information from complex relationships, often resulting in irrelevant retrieval results and problematic retrieval accuracy. Secondly, most of the existing methods do not correctly scale, generalize or customize with the nature of the intelligent maintenance domain. They are usually restricted to elementary search tasks with the use of text strings and then immediate subsequent composition work based on the initiating search stroke. This absence of domain-specific tuning is an inherent flaw of modern RAG techniques and renders them unable to respond to more complex numerous intelligent maintenance tasks and requirements that include high specificity and sensitivity in dealing with this problem. Furthermore, retrieved and generated content is not further processed and optimized in these systems, resulting in redundant information and incomplete answers [21]. The issues that arise from these demonstrate the need for further practicality and reliability improvements of RAG technology in order to meet the various requirements of industrial environments.

To mitigate the above challenges, the current study is guided by three research questions that indicate its research importance: First, what are the best ways to combine structured knowledge graphs with unstructured textual data to support fault reasoning, which makes use of both explicit relational information and implicit semantic information? Second, is lightweight fine-tuning of embedding models able to provide retrieval-accuracy improvements on par with more expensive full-LLM fine-tuning, and thus make domain adaptation in resource-constrained industrial environments possible? Third, in what ways can various specialized agents work together, via task decomposition, specific retrieval and refinement, to produce diagnostic accuracy that is better than that of monolithic reasoning systems? Such questions challenge the basic questions related to knowledge-representation heterogeneity, efficient domain adaptation, and collaborative reasoning architectures, and make the study relevant to a wider range of knowledge-intensive AI applications than fault diagnosis.

To address existing challenges, a knowledge-based collaborative multi-agent and knowledge graph framework (MAKG) is proposed for industrial equipment maintenance. By leveraging structured knowledge graphs and unstructured text data as a heterogeneous knowledge base, this framework provides the model with rich external knowledge and increases the framework's interpretability and traceability. Specifically, the framework uses a domain specific finetuned embedding model to map text to vector space more precisely, so that semantically similar texts are closer, improving the accuracy of retrieving relevant documents. The multi-agent module collaboratively processes problems for knowledge reasoning tasks. Task decomposition and multi path retrieval improve the accuracy of knowledge retrieval during the retrieval phase, and summary extraction and reflective iteration filter out irrelevant information to improve the precision of model responses in the generation phase.

The main contributions of this research are:

(1) The integration of structured knowledge graphs with RAG technology is proposed to leverage the associative properties of knowledge graph nodes, enabling the inference of more distant knowledge and thereby facilitating LLMs in exploring potential fault causes.
(2) A lightweight fine-tuning approach for embedding models is proposed to address the need for enhancing system retrieval accuracy, replacing traditional LLM fine-tuning methods while reducing costs for industrial applications.
(3) A plug-and-play multi-agent collaborative module is proposed to address the challenge of rapid integration into existing RAG-based intelligent maintenance systems. Multiple agents collaborate in planning and problem-solving, enhancing system flexibility and scalability while improving the accuracy of the framework's maintenance reasoning capability.

The rest of this paper is organized as follows: Section 2 reviews the latest research of LLMs and RAG in industry and Section 3 introduces the overall architecture of the MAKG system, detailing its components and the knowledge representation framework. Section 4 introduces an experimental study. Section 5 discusses and Section 6 concludes the paper.

2. Related works

Recent developments in intelligent maintenance and industrial diagnostics are the topic of data- and knowledge-based systems for smart factories [22]. These systems incorporate perception, cognition and decisions to address production problems in non-ferrous metal industries. Flow constraints are not considered in the distributed frameworks using space–time causal analysis for equipment maintenance in large processes involving sensor networks [23]. Digital twin-based and partial transfer learning-based service life diagnostic models address the domain shift in the maintenance monitoring of rotating machine faults [24]. These tools reduce downtime and increase resource utilization.

Based on these, structured knowledge is introduced into the industrial fault tasks via LLMs.

2.1. Recent advances of LLMs in industry

Recent advancements in Industry 4.0 and the growing prevalence of smart manufacturing have facilitated the integration of LLMs into fault detection, which is a critical component of industrial equipment operation and maintenance [25]. LLMs, distinguished by their robust natural language processing and multi-task learning capabilities [13], offer novel research perspectives and solutions for industrial fault detection.

In equipment maintenance, artificial intelligence and big data technologies have been widely applied. With the advancement of LLM technologies, an increasing number of studies are adopting knowledge-based fault diagnosis methods, utilizing text data to train and predict fault diagnosis models [7,8,10]. LLMs have been pre-trained on extensive corpora and have demonstrated proficiency in various knowledge reasoning tasks [26]. However, most publicly accessible LLMs are trained on general domain corpora (e.g., the Wikipedia corpus), which limits their applicability in specialized domain tasks. LLMs hold substantial promise in industrial applications. Previous research has shown that integrating domain-specific knowledge graphs into LLMs can enhance their performance in fault diagnosis, as evidenced by applications in aviation systems [13] and HVAC maintenance [14]. Additionally, models like CausalKGPT have improved shell manufacturing quality analysis by clarifying causal relationships through the integration of knowledge graphs and fine-tuning [27]. Meanwhile, multi-agent systems have gained increasing attention in fault diagnosis, which is reflected by a hybrid knowledge-based and data-driven approach for distributed fault diagnosis in heterogeneous multi-agent systems [28], and the implementation of LLM agents for automatic process regulation, demonstrating their ability to improve production processes and operational performance [28]. In industrial equipment maintenance, intelligent agent research has advanced into specialized domains with LLM integration. Latest works apply such agents to manufacturing process regulation and hydropower equipment inspection, enabling more adaptive fault detection and diagnosis compared to traditional rule-based counterparts [15,29].

The application of LLMs for diagnostic reasoning has demonstrated promising potential. However, this approach also encounters significant challenges, such as the hallucinatory nature of LLMs, which may result in errors within its outcomes, and the scarcity of domain-specific data, which is insufficient to meet the fine-tuning demands of LLMs [14]. Techniques like few-shot learning, data augmentation, fusion attention, and transfer learning have been explored to mitigate data scarcity issues [8,10,30]. To address these issues, RAG can be a potential tool to integrate external knowledge bases for improved reasoning.

2.2. RAG: Evolution and industrial potential

RAG offers advantages in complex industrial diagnostic tasks, enhancing knowledge timeliness and reducing hallucination through external data verification. Its traceability boosts credibility and minimizes frequent retraining, making it ideal for real-world applications. Despite these benefits, RAG's industrial deployment is limited, requiring further exploration.

The evolution of RAG technology has become increasingly sophisticated. Initially, RAG research commenced with simple annotation and entity retrieval [20], subsequently advancing to handle more complex structures such as text blocks and knowledge graphs, thereby enhancing retrieval granularity and data structuring. The introduction of advanced RAG and modular RAG signifies a new phase in RAG research. Advanced RAG overcomes early limitations through forward and backward retrieval strategies [31], while modular RAG allows for the replacement or reconfiguration of retrieval and generation modules based on specific application requirements, enhancing functional module practicality [32,33].

Further advancements in RAG technology include SELF-RAG, a self-reflective method that enhances generation quality and factual accuracy through specialized reflection tokens [34]. Graph RAG and LightRAG enhance the performance of traditional RAG and achieve a better understanding of semantic associations by integrating knowledge graphs into the retrieval process [35,36].

Although these RAG methods have demonstrated notable improvements in retrieval results and answer accuracy, their application in industrial maintenance is still in its infancy. In contrast to traditional methods using sensors or physical models, RAG systems can effectively handle the complexitieslarge-scalecale data and challenging environments [20,37]. However, the application of RAG technology in industrial equipment maintenance is still facing many challenges, including the general problem of low accuracy in industrial equipment maintenance and the complexity of processing the complex industrial data [21]. To address these challenges, this study proposed a method guided by fine-grained knowledge graphs and combined with lightweight fine-tuning embedding models and multi-agent modules to collaboratively extract subgraphs and document information more efficiently. The proposed approach can improve the accuracy of fault cause localization and problem resolution.

The literature review indicates three research gaps that were deemed to be critical and that warrant our research. To begin with, current RAG systems to diagnose faults use simplistic retrieval processes which view knowledge as a collection of flat texts thus not taking advantage of the rich relational representations found in industrial fault knowledge, and restricting the ability to have multi-hop logical reasoning about indirect causal effects. Second, existing methods are often based on either computationally intensive full-LLM fine-tuning which is not feasible in many industrial settings, or generic pre-trained models with no domain adaptation, which ignores the possible benefits of lightweight embedding fine-tuning of balancing performance and feasibility. Third, the vast majority of systems embrace monolithic reasoning architectures where one LLM deals with all aspects of query comprehension, retrieval, and generation, as such, it is not as modular and specialized as more advanced problem decomposition and refinement would require. The MAKG framework fills these gaps directly by integrating knowledge and graphs in a structured way, making it possible to perform relational reasoning, fine-tuning low-weight embeddings on the domain, and use a multi-agent cooperative architecture that allows addressing specific tasks and refining it in an iterative way.

3. Proposed method

The proposed framework, as illustrated in Fig. 1, achieves a synergistic enhancement through the integration of agent-assisted reasoning with knowledge graph-text fusion. Fault data from multi-source databases, including fault phenomena and troubleshooting methods, supports the creation of a knowledge base with semantically segmented text and domain-specific knowledge graphs. The two primary components of this framework, domain embedding fine-tuning and a multi-agent coordination retrieval model, are discussed in detail in Sections 3.1 and 3.2.

3.1. Domain embedding fine-tuning

As shown in Fig. 2, the embedding model is finetuned using LLM-generated question-answer pairs to improve retrieval accuracy in industrial equipment maintenance. The left side illustrates the pathway of embedding generation that can be implemented in which the domain-specific tokens are incorporated with input text and run through the pre-trained model to generate optimized embeddings. The right side introduces the contrastive learning model which learns the model by matching anchor queries with positive samples (important contexts) and negative samples (irrelevant contexts) and uses the embedding space to position similar semantically similar pairs nearer and different pairs further apart. Positive sample pairs are pairs of question-context, where the context is quite relevant to the question, directly helping to derive the answer. Negative sample pairs refer to question-context pairs that have low relevance. Domain tokens are industrial specific tags added to both queries and contexts to increase the awareness of domain entities in the embedding model so that there is a better match between the query and relevant context in the industrial equipment maintenance domain. Finally, the RAG system is integrated with the finetuned model to improve response precision for complex queries, and contrastive learning is used to improve the embedding space by learning to distinguish between correct and incorrect information.

3.1.1. Automatic generation of QA dataset

Question-answer pairs are automatically generated to guarantee that the training data is closely aligned with real equipment maintenance scenarios. The generated questions focus on key diagnostic elements such as symptoms, detection methods, and troubleshooting strategies. For each generated question q_i is paired with an answer a_i, which is either directly extracted from the equipment maintenance information or inferred from the KG (i.e. relationship) based on the question.

Let the input text be denoted as T, and the knowledge graph as KG. The LLM generates a set of questions $Q = (q_1, q_2, \ldots, q_n)$ based on the combination of the text and the entities in the KG, which can be expressed as:

$$Q = \mathrm{LLM}(T, \mathrm{KG}), \tag{1}$$

where LLM is a large language model based on the transformer.

For each question q_i, the corresponding answer a_i is extracted either directly from the text or inferred from the relationships within the KG. This forms the QA pairs $(q_1, a_1), (q_2, a_2), \ldots, (q_n, a_n)$. The quality and relevance of the generated question-answer pairs were checked by industry experts with many years of experience on industrial robot fault diagnosis along with checking their relevance to the core elements of the diagnosis, factual accuracy with industrial standards and technical documentation, and a practical application in real-world fault diagnosis cases.

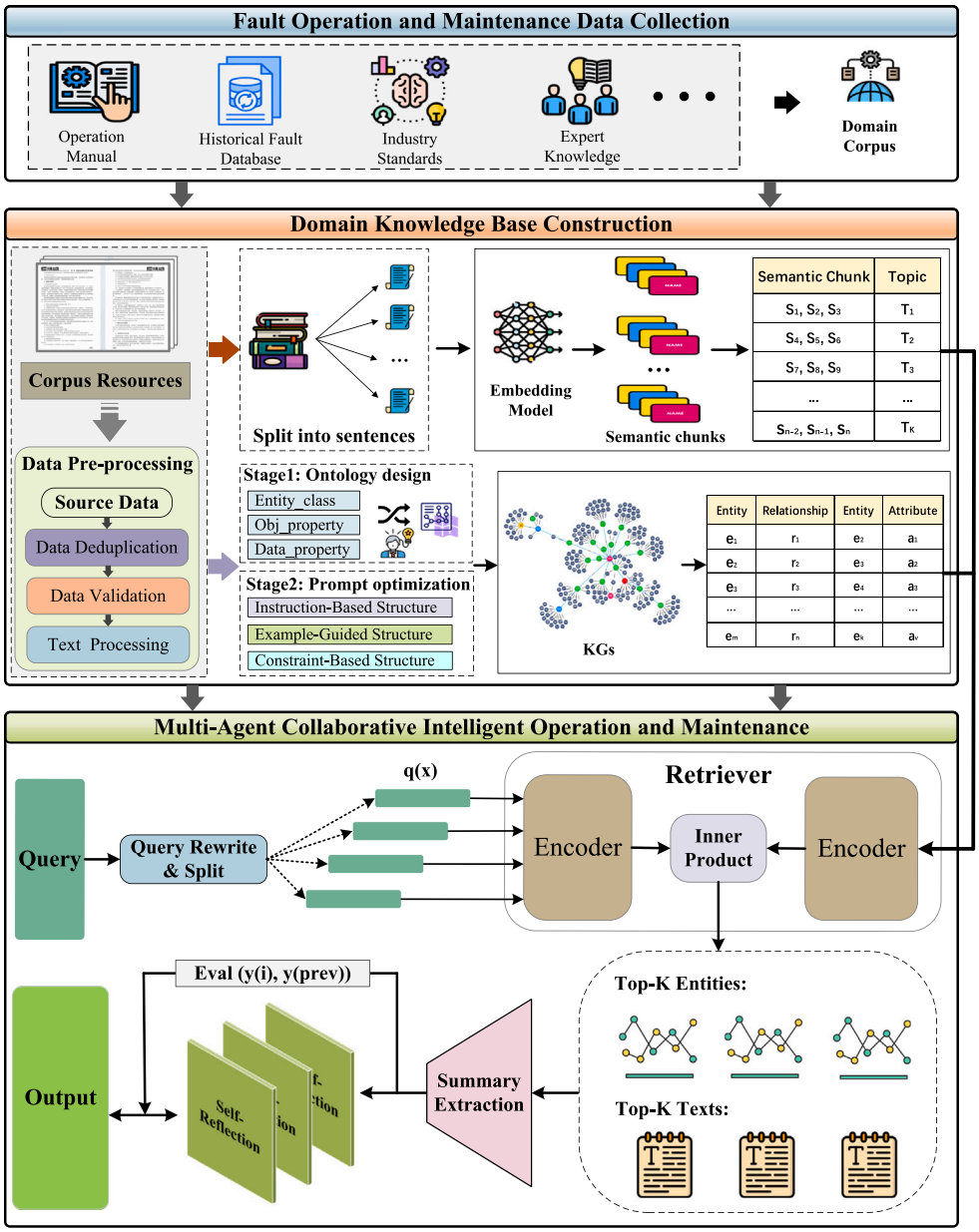

Fig. 1. Agent-KG integration framework.

In the future, this model architecture may be generalized to multimodal settings where images and videos are being used by substituting the text-only embedding model with the multimodal encoders that could handle heterogeneous datatypes into common representations spaces. Images of equipment, thermal images, or video frames showing the appearance of the fault would be passed through vision transformers or other such structures to identify visual features that would be matched to textual description via contrastive multimodal learning methods comparable to CLIP [38]. The generation of QA datasets would be extended to multimodal question-answer pairs with questions referring to visual observations and responses synthesizing visual and textual information. In the case of video inputs, dynamic fault behavior could be represented at the components of temporal modeling, including progressive vibration patterns or changing thermal distributions. Knowledge graph would need to be improved and added with the nodes representing visual concepts and time sequences, and the relations between them and textual descriptions of faults and diagnostic processes. Although this kind of multimodal extension is a significant future study beyond the present scope of the paper, by bringing up this direction of the pathway, we will have established our text-based base as a part of the eventual intelligent diagnostic systems.

3.1.2. Fine-tuning the embedding model

Once the QA dataset has been generated and validated, it is used to fine-tune the embedding model. The goal of fine-tuning is to optimize the model's ability to accurately represent both questions and answers in a vector space, allowing the system to retrieve the most relevant information for a given query.

The training process begins by embedding both the questions and answers into a continuous vector space. Let the embedding function be represented as $E(\cdot)$, where $E(q_i)$ and $E(a_i)$ represent the vector embeddings of question q_i and its corresponding answer a_i, respectively. The objective is to minimize the distance between $E(q_i)$ and $E(a_i)$, while maximizing the distance between $E(q_i)$ and any incorrect answer $E(a_j)$, where $i \neq j$ The contrastive loss function used for fine-tuning can be

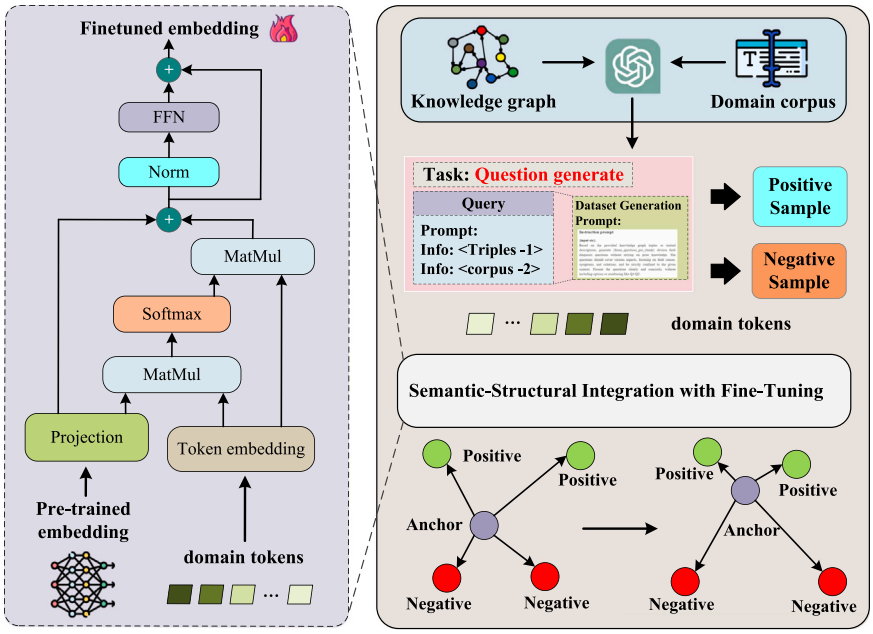

Fig. 2. Fine-tuning flowchart intelligent maintenance embedding model.

defined as:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp\left(\text{sim}(q_i, a_i^+)/\tau\right)}{\exp\left(\text{sim}(q_i, a_i^+)/\tau\right) + \sum_{j=1}^{K}\exp\left(\text{sim}(q_i, a_j^-)/\tau\right)}, \tag{2}$$

where:

- $\text{sim}(\cdot,\cdot)$: similarity function between query and answer embeddings,
- a_i^+: positive sample (correct answer) for query q_i,
- $\{a_j^-\}_{j=1}^K$: set of K negative samples (incorrect answers),
- τ: temperature parameter scaling the similarity scores,
- K: number of negative samples per positive pair,
- N: total number of training query-answer pairs.

3.1.3. Knowledge graph vectorization

For the effective utilization of knowledge graphs in industrial equipment maintenance, we adopt a Sentence-BERT-based embedding model, specifically acge_text_embedding, to encode each entity into a dense vector representation. This transformation involves several key steps:

(1) Entity Textualization: All the entities of the knowledge graph are transformed to a textual format. This can involve extraction of relevant attributes and descriptions of the entity in order to make sure that the textual output reflects the key characteristics of the entity.
(2) Text Embedding: These textualized entities are then represented in the form of vectors using a pre-trained embedding model that is fine-tuned on domain-specific question-answer datasets. In particular, the textualized entity is fed into the embedding function $E(\cdot)$ and produces the associated vector representation.
(3) Vector Storage: The generated vectors are stored into an efficient vector database in order to support fast retrieval and similarity calculations. In the process, the structured form of the knowledge graph is utilized and textual information is integrated, thus the system is improved in solving complicated queries.

The approach increases the precision of the retrieved information and enables a more competent processing of very specific and sensitive equipment maintenance.

3.2. Multi-agent coordination retrieval generation model

The proposed framework incorporates a multi-agent architecture to enhance the accuracy and effectiveness of equipment maintenance in industrial scenarios. The framework consists of four key modules that coordinate the retrieval and generation process to return precise and context aware responses. Fig. 3 is a schematic representation of this framework.

3.2.1. Query rewrite & split

The diagnostic process begins with query rewriting and splitting designed to address challenges posed by multi-hop questions and semantically ambiguous fault descriptions. Direct querying of such inputs often retrieves irrelevant information, degrading diagnostic accuracy. We employ an LLM-based approach operating in two sequential stages. First, the system performs semantic inference and completion to resolve ambiguities and enrich the query with implicit contextual information that experienced maintenance personnel would naturally understand but may not be explicitly stated in the original query. Second, the semantically enriched query undergoes systematic decomposition where complex diagnostic questions are broken down into focused subproblems, each targeting specific diagnostic aspects such as individual fault symptoms, particular equipment components, or distinct causal pathways. The fundamental purpose of this module is to maximize the number of condition-matching queries derivable from the original input. Consequently, the process generates multiple query variations and refinements, resulting in k sub-queries $q_1(x), q_2(x), \ldots, q_k(x)$, where k varies depending on the original question's complexity and ambiguity. This fine-grained decomposition enables more precise retrieval as each sub-query independently matches against highly relevant knowledge subgraphs and text fragments.

This approach involves:

(1) Decomposition into Sub-Problems: The original query Q is decomposed into k sub-queries $Q_1, Q_2, \ldots, Q_k$, each focusing on a specific subset of entities $e_1, e_2, \ldots, e_n$. This step is crucial for isolating distinct aspects of the equipment maintenance task.
(2) Entity Matching with Knowledge Graph Triads: The sub-queries are then matched against the system's vector repository, which is organized around knowledge graph triads. Each triad represents a discrete, relational piece of information, enabling precise entity matching.

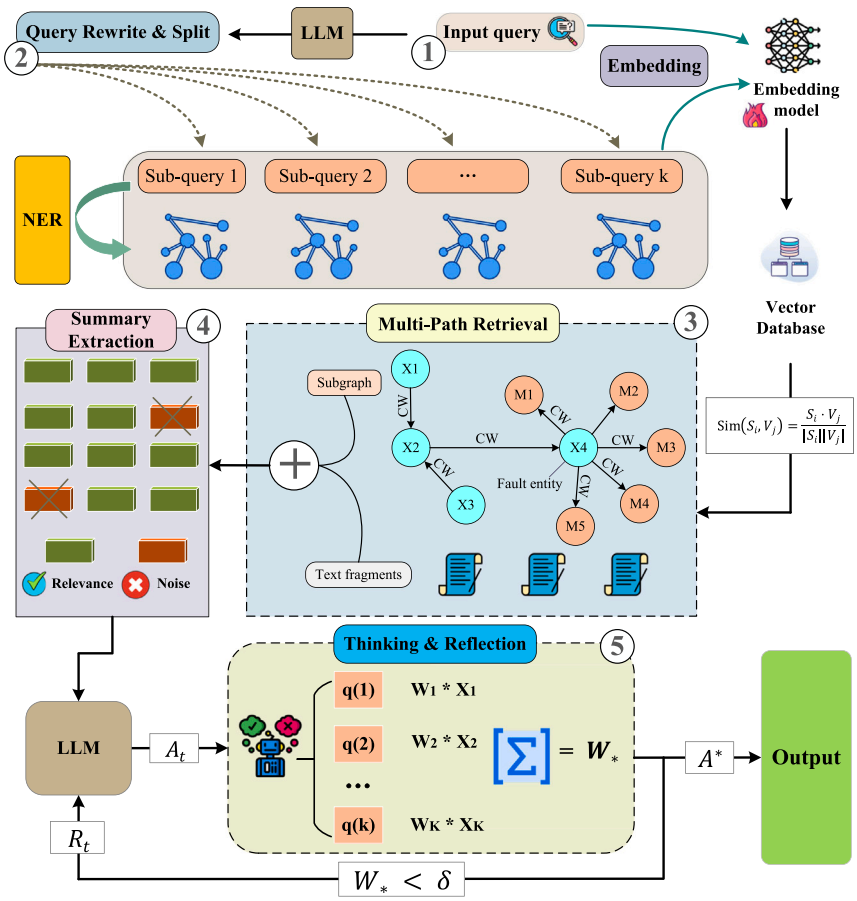

Fig. 3. Multi agent cooperative framework.

(3) Enhanced Retrieval: By breaking down complex diagnostic issues into smaller, more manageable components, the system retrieves more precise and varied knowledge.

Let the original query Q consist of n entities $e_1, e_2, \ldots, e_n$. The splitting agent transforms Q into sub-queries $Q_1, Q_2, \ldots, Q_k$, where each sub-query Q_i focuses on a specific subset of entities:

$$Q \rightarrow \{Q_1(e_1, e_2), Q_2(e_3, e_4), \ldots, Q_k(e_m)\}. \tag{3}$$

This transformation ensures that each sub-problem targets a specific aspect of the equipment maintenance task, leading to more focused retrieval from the knowledge base.

3.2.2. Multi-path retrieval

Once the problem is divided into sub-queries, the system invokes the Information Multiplex Recall module. This module performs similarity matching between the sub-queries and a pre-constructed vector knowledge base, which contains subgraphs from the knowledge graph and corresponding textual fragments.

As illustrated in Fig. 4, the retrieval of subgraphs involves matching key entities with high similarity and collecting neighboring nodes through adjacent edges E to construct the subgraphs. For each sub-query, the system retrieves the top-k subgraphs and text fragments most relevant to the fault problem. Let S_i represent the vector embedding of a sub-query Q_i, and let V_j represent the vector embedding of a subgraph or text fragment T_j. The similarity score $Sim(S_i, V_j)$ is computed using cosine similarity:

$$\mathrm{Sim}(S_i, V_j) = \frac{S_i \cdot V_j}{|S_i||V_j|}. \tag{4}$$

The top-k most similar subgraphs and corresponding text fragments $G_1, G_2, \ldots, G_k$ are then selected for each sub-query Q_i:

$$Top - k(Q_1) = \{G_1, G_2, \ldots, G_k\}, \tag{5}$$

where $Sim(S_i, V_j)$ is maximal.

3.2.3. Summary extraction

The Information Refining module processes retrieved subgraphs and text fragments by summarizing and removing noise, enhancing the precision of diagnostic data. This module introduces two key functions with clear definitions:

1. Relevance Function: Denotes the Relative Relevance of the retrieved subgraph or text fragment T_i, quantifying its contextual importance to the fault query. A higher value indicates stronger relevance to core diagnostic elements (e.g., fault causes, troubleshooting steps).
2. Noise Function: Represents the Relative Noise of T_i, quantifying the degree of contextual redundancy or irrelevance (e.g., duplicated content, off-topic information). A higher value indicates more interference with accurate diagnosis.

The module balances the relevance and noise of extracted segments, retaining only useful and non-redundant content. The summarization process is formulated as an optimization problem that maximizes total relevance while minimizing total noise:

$$\mathrm{Summarize}(T) = \arg\max_{S \subseteq T} \left(\sum_{T_i \in S} \mathrm{Relevance}(T_i) - \lambda \sum_{T_i \in S} \mathrm{Noise}(T_i) \right), \tag{6}$$

where λ is a regularization coefficient (set to 0.3 via experimental tuning) that balances the trade-off between relevance and noise, S denotes

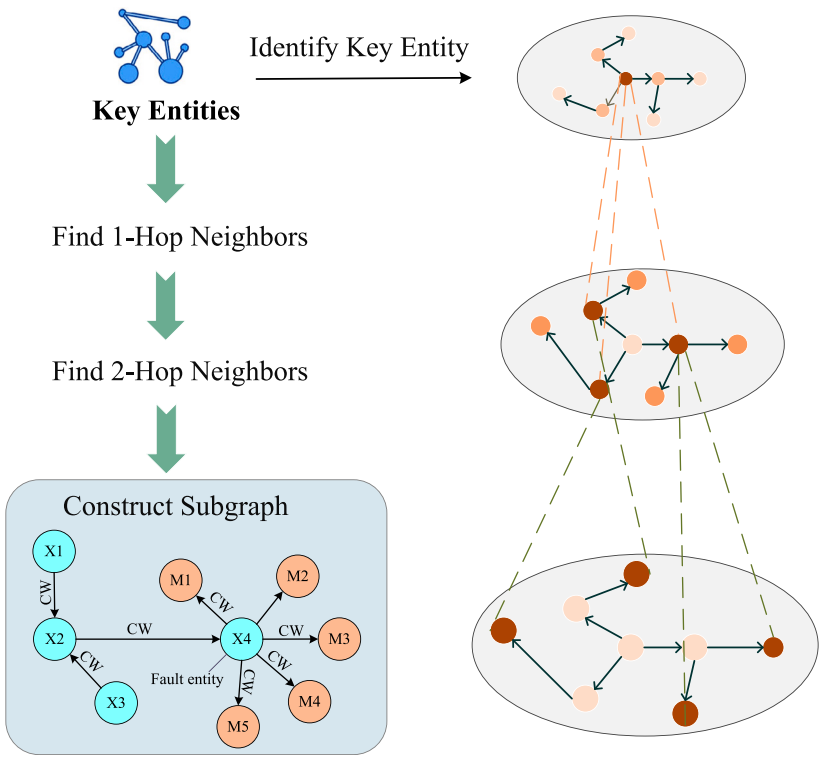

Fig. 4. Procedure of key entities mapping to knowledge graph sub-graphs.

the subset of retrieved fragments selected for summary generation, and T is the full set of retrieved subgraphs and text fragments.

3.2.4. Thinking & reflection

The self-reflection module iteratively refines generated answers, embodying the principles of Reflection, a framework that enhances language agents' decision-making through verbal reinforcement learning.

This module is based on its iterative self evaluation mechanism, which enables the model to iteratively update its answer accuracy assessment. It assigns an initial weight W to each of the K sub-questions, and uses an LLM as the evaluator to score each sub-question based on how well the response addresses it, thereby assessing the overall completeness of the answer and generating the final critique R. Then the scores are accumulated to see if they are more than a δ threshold. The model's responses are refined iteratively to increase accuracy based on the critiques R from the evaluation process as input if the scores do not exceed the threshold. This mechanism can identify shortcomings in each answer and incrementally refine the response.

Let A_t represent the answer at iteration t, and let R_t represent the reflection or critique of that answer. Let Q denote the query, and S_t represent the state of the model at iteration t. The state S_t encapsulates the model's understanding of the query and the current answer A_t. The refined answer A_{t+1} is computed as:

$$A_{t+1} = A_t + \lambda f(S_t, Q), \tag{7}$$

$$S_{t+1} = g(S_t, A_{t+1}), \tag{8}$$

where $\lambda \in [0, 1]$ is the reflection weight, balancing the influence the revised answer, f is a function that generates the critique based on the current state and the query, g is a function that updates the state based on the refined answer A_{t+1}. This iterative mechanism gradually optimizes the response, converging toward an optimal answer A^*, which is more contextually precise and aligned with diagnostic requirements.

3.3. Agent-KG integration framework

Fig. 5 shows how the fault knowledge base of the MAKG system allows it to resolve industrial robot faults and locate faults precisely. Before applying the multi-agent coordinated intelligent maintenance framework, each agent module is initialized to perform its specific function: the Problem Rewriting and Splitting agent divides the input fault query into sub problems for enhanced retrieval efficiency, the Information Multiplex Recall agent retrieves relevant sub graphs and text fragments from the vectorized Knowledge Graph and data sources, the Information Refining agent cleans and condenses the results to maximize data usefulness, and the Self Reflection agent iteratively improves the answer through multiple cycles of self assessment to guarantee precise diagnostic answers and corresponding knowledge sources, as outlined in Algorithm 1.

To prove the "plug-and-play" characteristic of the multi-agent module, we use the situation of integrating a pre-packaged "Fault Code Interpretation Agent" into the original framework (including Query Decomposition, KG Retrieval, Information Filtering and Reflection Iteration Agents) as an example. The whole process is end-to-end and does not require any changes to the codes: the pre-packaged Fault Code Interpretation Agent is directly added to the framework, no adjustments are required to existing agents. The built-in adaptive message router of the framework automatically recognizes the input/output interfaces of the new agent and establishes communication links. When processing a query such as "Explain fault code F023 and its solution" the framework first calls the Query Decomposition Agent to separate the task into "interpret fault code F023" (assigned to the new Fault Code Interpretation Agent) and "retrieve maintenance solutions for F023" (assigned to the KG Retrieval Agent). The Information Filtering Agent then screens valid results from both paths and Reflection Iteration Agent integrates the results (e.g. "F023 indicates power supply instability" from the new agent and "check voltage regulator and reset system" from KG) to generate the final response. This case validates that new pre-packaged agents can be easily plugged into the framework with just input given, fully implementing the "plug-and-play" nature of the multi-agent module.

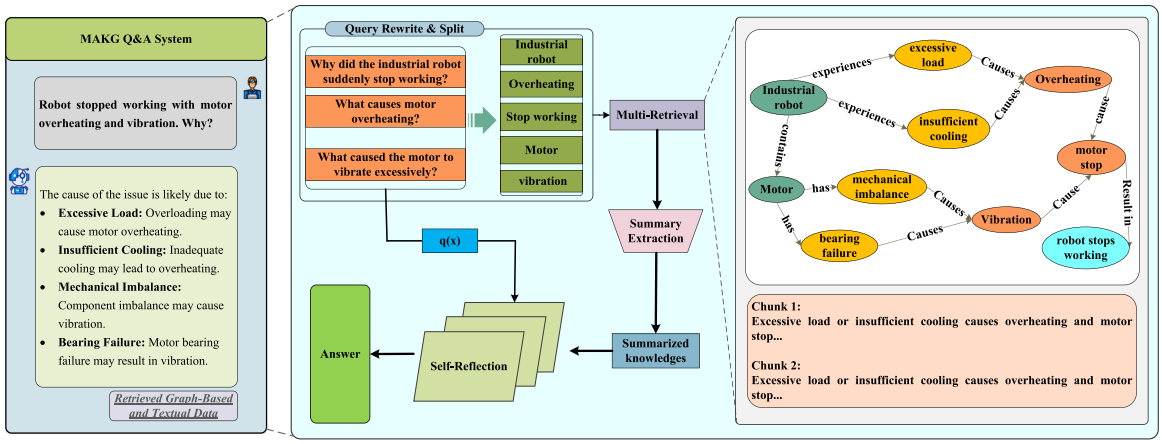

Fig. 5. Framework application example diagram in industrial scenarios.

Algorithm 1 Multi-Agent Coordination Retrieval and Generation Algorithm

Input: Query q, Embedding model M, Vector DB D, Max iter $I_{\max}$, Threshold ϵ
Output: Final diagnosis y, Relevant sources S

Step 1: Sub-Query Decomposition and Retrieval
Split q into sub-queries $Q = \{Q_1, Q_2, \ldots, Q_k\}$
Retrieve top-k subgraphs t_i and fragments f_i
Store pairs (t_i, f_i) in retrieval set R

Step 2: Information Refinement
for each $(t_i, f_i) \in R$ **do**
    Refine t_i and f_i
end for

Step 3: Iterative Self-Reflection
Initialize $y_{prev} \leftarrow \emptyset$, $\delta \leftarrow \infty$, $i \leftarrow 0$
while $i < I_{\max}$ and $\delta > \epsilon$ **do**
    $y_i \leftarrow$ Self-Reflection$(refined(t), refined(f))$
    $\delta \leftarrow$ LLM-Eval(y_i, y_{prev})
    $y_{prev} \leftarrow y_i$, $i \leftarrow i + 1$
end while

Step 4: Return Result
return $y \leftarrow y_i$, $S \leftarrow R$

4. Experimental study

In order to evaluate the performance of the proposed MAKG intelligent maintenance framework, we conducted empirical evaluations on a comprehensive dataset encompassing real-world intelligent maintenance task queries, as well as on a publicly available dataset specifically tailored for assessing systematic reading comprehension and question-answering capabilities. Subsequently, we quantitatively compared the performance metrics of our methodology against those of several well-established RAG approaches, thereby elucidating the scientific merits and limitations of our proposed framework within the context of intelligent maintenance tasks.

4.1. Dataset description

To fully validate the efficacy of the framework, we built an industrial robot fault diagnosis dataset, IFD-QA, which includes fault-related question-answer pairs. The data set is based on the records of fault case for enterprises, including 210 fault cases of industrial robots in various production environments and operation states. These scenarios are distributed evenly between 7 core fault categories (each of which accounts for 13%–16% of the total), to ensure balanced data distribution. Real-world maintenance logs and technical documents from industry partners were used to collect data. Fault questions were reviewed and standardized with the help of an expert consultation; answers were synthesized based on diagnostic procedures and solutions described in official resources. Especially, each of the scenarios is labeled with elaborate tags such as fault type, cause, and the components affected. Preprocessing of text consisted of tokenization, lowercasing, and removal of redundant symbols without losing important technical terms and numerical values. The division of the dataset is detailed in Table A.1. These scenarios are categorized into three distinct types:

(1) **Simple:** Questions are straightforward and answers can be directly found in the context;
(2) **Multi-Context:** Answers require information from multiple related contexts or text blocks, testing the system's ability to integrate and correlate diverse information sources to form complete answers;
(3) **Reasoning:** Answers necessitate logical inference from the context, thereby evaluating the system's reasoning and deep comprehension capabilities.

Table 1
Comparison of retrieval performance between base and fine-tuned embedding models.

Evaluation metric	Base model	Fine-tuned model
Precision@10	0.8115	0.9183
Recall@10	0.7527	0.8891
Hit Rate@10	0.7834	0.9268

4.2. Experimental setup

The experimental setup for this study was conducted on an Ubuntu 20.04 LTS system with an NVIDIA RTX A800 GPU equipped with 80 GB of GPU memory. We used the Torch 2.3.1 framework for fine tuning the embedding model and trained for 10 epochs. The batch size was set to 8, and the learning rate was configured at 2×10^{-5}. A gradient accumulation strategy was implemented, accumulating gradients every 2 steps. The fine-tuning dataset contained approximately 4000 entries, typical for embedding models given their small parameter size. Training the lightweight fine-tuned embeddings (acge text embeddings, 12M parameters) took 45 min on a single NVIDIA RTX A800 with a

maximum of 5 GB GPU memory used. For comparison, a full fine-tuning of a 7B-parameter LLM will require 38 GB of GPU memory and more than 5 h. As illustrated in Table 1, this lightweight method is able to provide impressive performance boosts. These results validate the approach that has low computational cost and high retrieval efficiency.

The large language base model used in this study is DeepSeek-v2 16B [39]. It has a reasonable performance to computational efficiency ratio, and the 16B parameters can support complex fault diagnostic logic, but can be implemented on standard hardware. It is also knowledge-intensive with reasoning capabilities that are strong, it is very competent in the context of our multi-agent system, it is very much integrated into the retrieval-augmented generation process and it is readily available and with good licensing to research and industry. An industrial machinery Fault Diagnosis Knowledge Graphs (FDKGs), KDKG, was constructed using enterprise fault data to assess the model's effectiveness and the impact of different knowledge graphs on fault diagnosis tasks.

To thoroughly assess the model's effectiveness and the impact of different knowledge graphs on fault diagnosis tasks, two distinct industrial machinery Fault Diagnosis Knowledge Graphs (FDKGs) were constructed. These FDKGs, designated as KDKG and FACKG, were develo ped using fault data provided by the enterprise. The specific distinctions between these knowledge graphs are outlined in Table A.2. To quantify the performance of the MAKG framework, we employ several evaluation metrics: faithfulness, context recall, answer correctness, context precision and time consumption. To ensure consistency and reliability, we apply a unified prompt template across all evaluations and perform cross-evaluation using multiple LLMs with different architectures and training data. The experiments were repeated five times and take the mean value as the final output. The formulas for these metrics are delineated as follows:

$$F = \frac{N_c}{N_t}, \tag{9}$$

$$CR = \frac{N_{c,GT}}{N_{GT}}, \tag{10}$$

where:

- F: Faithfulness score, measuring the proportion of accurate diagnoses in the generated response.
- N_c: Number of correct diagnoses extracted from the response.
- N_t: Total number of diagnoses in the response.
- CR: Context recall, quantifying the retrieval of ground truth (GT) diagnoses from the provided context.
- $N_{c,GT}$: Number of correctly identified GT diagnoses from the context.
- N_{GT}: Total number of GT diagnoses available in the context.

$$AC = \omega_{AS} AS + \omega_F \frac{|V_{at}|}{|S_a|}, \tag{11}$$

where:

1. AC: Answer Correctness, a composite score assessing both semantic fidelity and factual grounding in the generated response.
2. AS represents answer similarity, a measure of the semantic similarity between the generated answer and the true answer.
3. ω_{AS} is the weight assigned to answer similarity.
4. $|S_a|$ denotes the total number of extracted points in the answer.
5. LLM is utilized to assess whether the extracted points can be inferred from the known truth set (Truths). The number of points that can be inferred is represented by $|V_a t|$.
6. ω_F is the weight assigned to fact relevance.

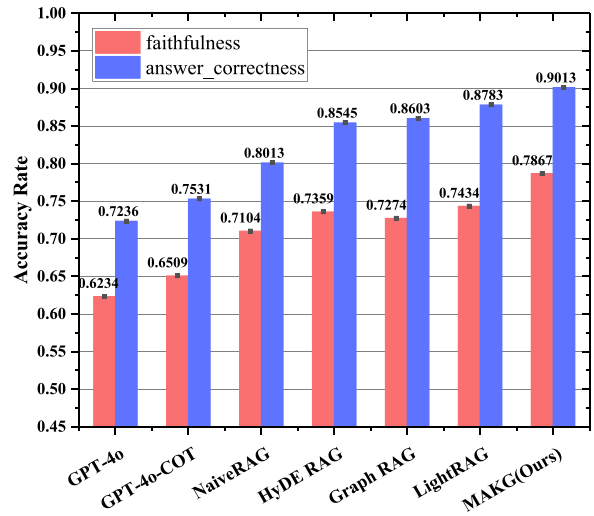

Fig. 6. Comparison of accuracy rates of different models.

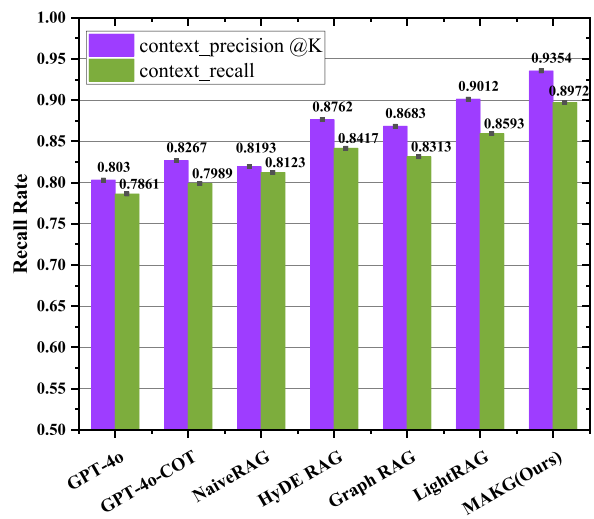

Fig. 7. Comparison of recall rates of different models.

Both $\omega_A S$ and ω_F are adjustable weight parameters, each set to 0.5.

$$CP_K = \frac{\sum_{k=1}^{K}(P_k \times v_k)}{N_{correct,topK}}, \tag{12}$$

$$P_k = \frac{TP_k}{TP_k + FP_k}, \tag{13}$$

where:

- CP_K: Context Precision at rank K, evaluating the weighted precision of relevant chunks in the top K retrieved results.
- P_k: Precision at rank k, measuring the ratio of true positives among retrieved items up to rank k.
- TP_k: True positives at rank k, the number of correctly identified diagnoses in the top k results.
- FP_k: False positives at rank k, the number of incorrectly identified diagnoses in the top k results.
- $v_k \in \{0, 1\}$: Binary relevance indicator at rank k (1 if the chunk at rank k is relevant, 0 otherwise).
- $N_{correct,topK}$: Total number of correct diagnoses in the top K results.

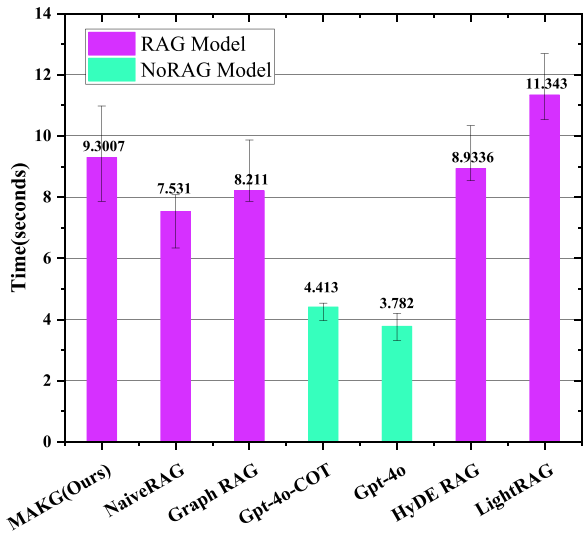

Fig. 8. Comparison of time consumption across different models.

4.3. Baselines

The efficacy of the MAKG model in the context of intelligent maintenance is systematically assessed through a comparative evaluation against several RAG baseline methodologies tailored to intelligent maintenance tasks. This comparative analysis is designed to substantiate the capability of the MAKG model to accurately diagnose faults, thereby highlighting its comparative strengths and limitations in relation to existing RAG intelligent maintenance approaches within this domain.

1. Graph RAG [35] constructs a knowledge graph, partitions it into communities for efficient summarization, and aggregates partial responses for complete answers, suitable for intelligent maintenance comparisons.
2. HyDE RAG significantly enhances LLM outputs. It innovatively employs Hypothetical Document Embedding (HyDE) [31], generating plausible text based on queries. Then, LLM reranking is utilized to reassess the initially retrieved documents, refining their relevance for more accurate results.
3. LightRAG makes RAG simple [36]. The indexing mechanism employed is lightweight and reduces computational costs in knowledge retrieval. It allows for faster discovery of relevant documents in large knowledge bases. Additionally, LightRAG optimizes the combination of retrieved data and generation model to generate answers in an efficient and context - aware manner.
4. Comparative experimental tests were conducted using a naive RAG (classic RAG without special processing) and non-RAG pure LLM reasoning methods, including GPT-4o and GPT-4o's question-answering method based on Chain of Thought (COT) [40] reasoning.

4.4. Experimental results

4.4.1. Results of baselines comparison

The comparative experiment depicted in Figs. 6, 7 and 8 demonstrates that the MAKG method significantly outperforms other approaches on the IFD-QA dataset. Non-RAG methods like GPT-4o and GPT-4o-COT had the lowest recall and precision rates. Among RAG-based methods, NaiveRAG performed the worst. HyDE RAG improved upon NaiveRAG, achieving higher recall and precision rates. Graph RAG and Light RAG, on the other hand, utilize knowledge graphs as their knowledge base. Graph RAG showed improvements in accuracy, while Light RAG, an enhanced version of Graph RAG, outperformed Graph RAG in both recall and precision. MAKG outperformed all methods in recall and precision. Non-RAG methods had faster response times, with Naive RAG being the quickest among RAG methods. Light RAG consumed significantly more time, while Graph RAG, HyDE RAG, and MAKG had similar time consumption. However, MAKG exhibited greater variability in time usage. Fig. 9 illustrates the impact of different top-k parameter settings on model performance. The recall rate reaches a plateau and shows no significant improvement beyond k=10, while the accuracy rate begins to slight decline when k exceeds 10.

4.4.2. Results of ablation experiment

In the ablation experiment in Fig. 10, it is evident that incorporation of optimized KACKG and fined-tuned embedding models have a major impact on recall and precision. There are also metrics that are affected by the query rewrite and split module. The lack of the retrieval content summary module and the self-reflection module cause an important effect on the accuracy, whereas the influence on the recall is rather minimal. The system modules are connected with each other and operate in a synergistic manner. The query rewrite and split module come up with focused sub-queries, and this facilitates the multi-path retrieval module in searching efficiently. Embedding model fine-tuning increases retrieval accuracy and the knowledge retrieved is directly utilized in summary extraction and self-reflection modules and therefore, this leads to better system performance. In the case of the self-reflection module, the retrieved knowledge can be used to act as a factual anchor to test the generated information, which will help correct the misjudgments, including wrong cause of fault associations. This is found to reduce reflection errors by 5.2% and enhance a better diagnosis accuracy by 5.9 percent than when there is no reflection with retrieved knowledge.

It is important to note that the KACKG removal causes the most significant decrease in performance when compared to all the ablated modules. Being a fine-grained fault diagnosis knowledge graph with 2869 entities and 6217 triples, KACKG explicitly represents the structured relationships of fault symptoms, root causes, affected components, and remediation solutions relationships that cannot be fully expressed in unstructured text. In the absence of KACKG, the system becomes incapable of doing multi-hop reasoning, which is essential to diagnosing complex faults in which the relationship between observed symptoms and cause is indirect.

We acknowledge this fact and realize that our first ablation analysis can be further improved with more detailed elaboration of the mechanics. The performance drop observed in the case of the elimination of every module is ascribed to the functional contribution of each to the diagnostic pipeline. When the fine-tuned embedding model is eliminated and replaced by the base model, then the system cannot effectively project domain-specific vocabulary and fault specifications to the vector space, so the queries and the relevant knowledge are incompatible. This has direct adverse effect on retrieval precision, which is translated to lower overall accuracy. The absence of the query rewrite and split module would mean that the system would handle complex multi-faceted fault queries as a single query, which would more frequently than not give widely general or half-relevant information rather than the exact subgraph needed by each particular aspect of the fault. The summary extraction module serves as an important filter to eliminate redundant and tangentially-related information of the information that has been retrieved. Without it, the language model is inputted with noise and redundant or uninformative information which waters down meaningful signal and results in less diagnostic responses. Finally, the self-reflection module allows refining the answers by generating them repeatedly, comparing them with the already stored facts, and detecting logical inconsistencies or incomplete thoughts. This is because its lack causes the system not to have a mechanism of rectifying underlying diagnostic errors early enough, especially in complicated cases that require multi-step reasoning.

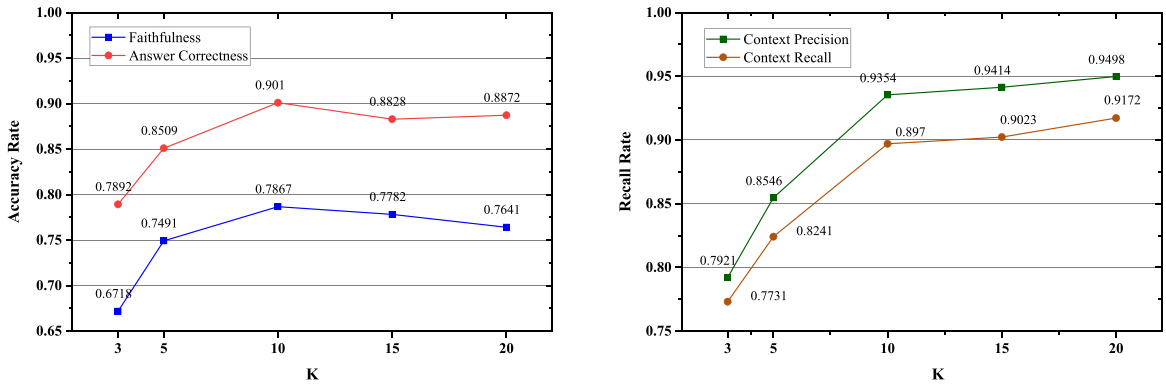

Fig. 9. Result of Top-K performance on IFD-QA.

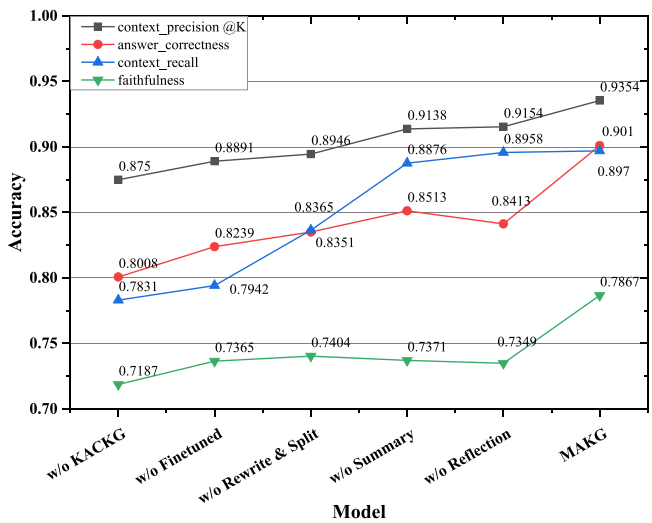

Fig. 10. Plots of ablation experiments regarding different modules.

5. Discussion

The examination of experimental outcomes reveals that GPT-4o and GPT-4o-COT, using non-RAG methods, perform sub optimally due to inability to fine-tune domain knowledge. Among text-based RAG methods, HyDE RAG outperforms Naive RAG, benefiting from its use of LLM for knowledge reorganization and optimization. However, both methods suffer from limited retrieval approaches and fail to effectively address the long-tail distribution of knowledge, resulting in a lack of critical information and thus only marginal performance improvements.

Among KG-based RAG methods, Graph RAG achieves relatively high accuracy due to its use of hierarchical community clustering and customized summarization, but its recall rate is lower because its knowledge graph represents entities and relationships in a simplistic manner, failing to capture implicit fault relationships. In contrast, Light RAG improves upon Graph RAG by integrating knowledge graph retrieval into a hierarchical retrieval approach, significantly enhancing both accuracy and recall rates compared to Graph RAG. Non-RAG methods are faster than RAG due to the absence of retrieval steps. Although our method incurs increased system response time due to additional processing, it remains within an acceptable range.

Experimental results demonstrate that the proposed MAKG method, which utilizes structured knowledge graphs and unstructured text as a knowledge base along with a multi-agent collaborative strategy, effectively leverages the relational edges in the graph to discover semantic relationships and infer indirect connections, outperforming other methods in handling complex maintenance processes. It also features a more precise recall mechanism and advanced fault analysis techniques, which contribute to its superior performance in handling complex maintenance processes compared to other methods. In comparison to single-agent methods like HyDE RAG, MAKG shows definite superiority in dealing with complicated maintenance problems. We ascribe this edge to the cooperative division of work among numerous agents that successfully addresses the limitations of LLMs in handling long contexts whilst significantly improving granularity and accuracy of responses. Through spreading tasks among various agents, MAKG enables specialized roles, thus improving the efficiency and accuracy of fault diagnosis in complex systems.

Results from the ablation study show that the fine grainefine-graineda significant impact on experimental outcomes and guides system reasoning. In particular, the fine-tuning embedding model and query rewrite & split module significantly improve the system recall rate by directly affecting information retrieval. On the other hand, the information summary and reflection iteration modules play a major role in determining the answer accuracy, requiring a finer processing of retrieved data. Thus, these modules are critical for accurate fault diagnosis. In experiments with different top-k settings, it was observed that increasing k to retrieve more information does not necessarily improve model performance. This is because excessive retrieved content as context can hinder the LLM's knowledge analysis and reasoning capabilities, leading to a decline in model performance. Therefore, it is crucial to balance recall and accuracy by selecting an appropriate k value. Experimental results indicate that $k = 10$ yields the best model performance.

MAKG has been confirmed by preliminary industrial deployments and has shown not only the increase in diagnostic accuracy but also the troubleshooting efficiency. The industrial practitioners normally need about one minute to detect the fault locations and suggest corrective measures, whereas detecting complex faults can take up to two minutes. Comparatively, on a corpus of 210 real world industrial scenarios, MAKG achieved an average response time of 9.3007 s which highlights a significant practical value and efficiency.

5.1. Managerial implications

The results of our work can be used by organizations considering implementing AI-assisted maintenance. First, in terms of resource feasibility, the lightweight embedding method achieved a high accuracy of more than 90% and introduced a small set of computational requirements (less than 45 min of training on a single GPU, using a memory of 5 GB, compared to full LLM fine-tuning that needs 38 GB of memory and over five hours). This proves that intelligent diagnostics can be introduced by even resource-limited manufacturers without the use of prohibitive infrastructure investment. Second, concerning knowledge management, the ablation study revealing that KACKG

removal decreased accuracy from 90.1% to 80.08% indicates that systematic structuring of maintenance records and expert knowledge yields substantial operational value. Organizations therefore ought to treat knowledge curation as an investment venture just as much as technology acquisition is treated. Third, under deployment strategy, our modular architecture will give us the opportunity to add its basic functionality gradually starting with simple scenarios of diagnostic cases and then moving onto more complex situations, which will decrease the risk of implementation and add organizational trust. Fourth, in terms of performance expectations, 9.3 s of the average response time is an approximate tenfold improvement of manual diagnosis but still slower than less complex automated diagnostic technologies, which make managers carefully consider the technology capabilities and how they are applied in the context of an operation where overall accuracy warrants the response time.

5.2. Limitations and future directions

The proposed methodology demonstrates several strengths, but also faces important limitations that require systematic consideration and point toward a coherent research roadmap for improvement. The closest obstacle is system responsiveness, in which the average response time of 9.3 s of the framework, even though significantly shorter than the manual diagnosis process that takes one to two minutes, is still not as desirable as the relatively simple procedures. The root cause of this latency is that the architectural design places diagnostic accuracy in the multi-agent coordination and which refines itself with the cumulative processing overhead of sequential communication between agents and knowledge graph traversal operations. Overcoming this efficiency problem is the basis of further enhancements as we will introduce asynchronous agent communication protocols that allow parallel implementation of decomposed sub-queries instead of sequential execution and intelligent caching of commonly accessed knowledge subgraphs that initial analysis indicates would speed up about thirty percent of common fault pattern queries. Such coordinated optimizations aim at improving response time to five to six seconds with diagnostic quality, which is a forty to forty-five percent improvement that can provide the computational headroom to more advanced capabilities.

The current text-only processing of the framework is based on this efficiency foundation and is a critical capability boundary that restricts the applicability to the entire range of multimodal industrial diagnostic conditions based on thermal imaging, vibration waveforms, acoustic signatures, and visual inspection data. To overcome this weakness, one will need to design domain-specific multimodal embedding models that produce shared representation spaces in which semantically related information across modalities can be efficiently compared and retrieved, and that must be developed based on the efficiency gains of the optimized retrieval infrastructure. We initiate our staged extension plan with the addition of simpler modalities (e.g., the images of static equipment) and continue to add more complex time-series sensor modalities, where we realize that each modality comes with new computational costs and integration challenges with various industrial hardware that need to be systematically verified. Moreover, the factual validity of the framework is very high due to its basis on retrieved knowledge, but there can be some lack of diversity in generation when retrieved information highly limits the solution space, possibly missing new patterns of faults not observed in history. Future research will examine methods to implement a balance between knowledge grounding and creative hypothesis generation, such that the system is able to reason outside of the carefully documented cases and yet be reliable.

Such technical advances are ultimately useful in the larger purpose of practical industrial implementation, where actual operational contexts provide challenges that are not limited to laboratory validation. Interoperability with existing heterogeneous industrial systems needs to support different data formats, different communication protocols, and different security requirements, and the reliance of the framework on complete knowledge graphs introduces continuous maintenance needs as equipment changes and new fault modes arise through experience of operation. In order to overcome these deployment issues and use the efficiency and capability improvements discussed above to advantageous effect, we will devise edge computing deployment approaches that decentralize inference capabilities to local controllers to achieve low-latency response and remain connected to centralized knowledge repositories to allow incremental knowledge graph updating mechanisms that can absorb new fault cases without retraining the entire system using automated consistency validation. Temporal graph neural networks provide potential solutions to the problem of prediction of changing fault patterns in streaming sensor data, and initial validation in CNC machining settings has shown that these solutions can give early indicator degradation warnings several days before standard threshold-based alarms can be raised. Most importantly, the need to achieve diagnostic reliability in deployment situations in which erroneous conclusions may lead to inappropriate maintenance actions or safety risks necessitates the adoption of ensemble verification strategies that produce hypothesis by following multiple reasoning paths in order to measure consistency, and confidence scoring mechanisms that quantify diagnostic confidence and identify situations with low confidence to be subject to human expert scrutiny as opposed to offering possibly unreliable diagnoses. This combined strategy of deployment reliability is a logical continuation of the underlying efficiency gains and the longer capabilities, forming a logical line of passage.

The MAKG framework is currently working with the static datasets however the industrial application will involve real time adaptation to the streaming sensor data and changing conditions. This will be tackled in future by introducing edge computing to dynamically synchronize the knowledge graph so that the knowledge graphs can be updated with incremental entity and relation updates without the need to retrain the model. The temporal graph neural networks will identify malpractices in the data streams, which will result in selective fine-tuning of the embedding models. These will be improvements that have been tested by field trials in CNC settings and will guarantee low-latency responsiveness and continuous performance in manufacturing machineries.

Furthermore, the impact of model-induced hallucinations on the reliability of assessment outcomes must be addressed, particularly when leveraging an LLM as an evaluator.

6. Conclusion

In industrial equipment maintenance, an auxiliary knowledge reasoning framework has been developed to harness the synergistic benefits of professional domain text, KG, and LLM through a multi-agent approach. This work aims to improve how knowledge is retrieved and reasoned during maintenance, especially in complex industrial settings where information can be scattered and hard to process. We incorporate a detailed knowledge graph along with semantically structured text as supplementary information, which helps speed up and improve the accuracy of fault localization in question answering. A lightweight model is also introduced to enhance retrieval without the need to fully fine-tune the large language model, which not only saves computing resources but also maintains high performance. Using multiple LLM agents together allows for more effective problem solving, better search results, deeper cause analysis, and more accurate fault identification. MAKG is built as a knowledge search engine that helps fill the gaps often found in existing industrial knowledge bases, making it easier to find relevant diagnostic information. Our experiments show that MAKG performs better than other methods in diagnosing faults, and removing any part of the framework leads to a noticeable drop in performance. Looking ahead, we plan to explore how visual data can be included to test the model across different types of inputs. We also aim to make the system faster by improving its structure.

Table A.1
Datasets partitioning settings.

Datasets	Simple	Multi context	Reasoning	Num
IFD-QA	0.5	0.3	0.2	210

Table A.2
Statistics of the KGS.

FDKGs	Entities	Relations	Triples
KDKG	1587	8	2643
KACKG	2869	14	6217

CRediT authorship contribution statement

Jia Lin: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chong Chen:** Writing – original draft, Methodology, Formal analysis, Data curation. **Tao Wang:** Writing – review & editing, Supervision, Software, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Zhuowei Wang:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Data curation. **Lianglun Cheng:** Writing – review & editing, Validation, Supervision, Resources, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Supplementary Experimental Datasets and Detailed Results

Table A.1 shows the specific partitioning structure of the IFD-QA dataset that we used in our experimental validation. The data set will consist of 210 fault diagnosis scenarios that are spread over three levels of difficulty, thus allowing a thorough evaluation of the diagnostic capabilities of the framework. Simple questions take up half of the data and entail simple cases of fault in which answers can be found directly in single contextual sources. Multi-Context questions include 30% of the cases and require the combination of the information in more than one related block of text or knowledge-graph entity, hence, challenging the ability of the system to correlate different sources of information. The rest 20% is reasoning questions that require logical inference over and above what is stated, and assesses the logical understanding and deduction abilities of the framework. This uniform distribution is a guarantee of a strong evaluation of different diagnostic complexities in the real-life industrial maintenance situations.

Table A.2 gives detailed statistics of the two Fault Diagnosis Knowledge Graphs (FDKGs) built and used in this paper. The Knowledge-Driven Knowledge Graph (KDKG) consists of 1587 entities that are linked by eight different types of relations, constituting 2643 triples that represent basic knowledge of fault-diagnosis, such as equipment parts, fault symptoms and simple causality. The Knowledge-Augmented Comprehensive Knowledge Graph (KACKG) is a finer-grained knowledge structure, consisting of 2869 entities, 14 relation types, and 6217 triples, and includes detailed diagnostic processes, maintenance solutions, and complex multi-hop causal paths. The large disparities in scale and granularity between these two knowledge graphs enable an ablation analysis to determine how much of the knowledge comprehensiveness affects the diagnostic accuracy, which is addressed in Section 4.4.2. The two knowledge graphs were built using enterprise fault records, maintenance logs, and expert knowledge, which was proven by industry professionals.

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